METRIC

MIL-DTL-24494C(SH)
22 June 2018

SUPERSEDING
MIL-T-24494B(SH)
19 October 1988

DETAIL SPECIFICATION

TETRASODIUM, TRISODIUM, AND DISODIUM SALTS OF ETHYLENEDIAMINETETRAACETIC ACID WATER TREATMENT CHEMICALS, BOILER, SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tetrasodium, trisodium, and disodium salts of ethylenediaminetetraacetic acid (EDTA) water treatment chemicals used in boilers for shipboard use.

1.2 Classification. EDTA is of the following types, as specified (see 6.2):

a. Type I – Tetrasodium salt

b. Type II – Trisodium salt

c. Type III – Disodium salt

1.3 Part or identifying number (PIN). PINs to be used for the sodium salts of EDTA acquired to this specification are created as follows:

M	**24494**	**-**	**X**	**-**	**XXX**
Prefix for Military Specification	Specification Number	-	Type (see 1.2)	-	Unit of Issue in Pounds
Examples: M24494-III-001, M24494-II-025					

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN CHEMICAL SOCIETY (ACS)

ACS Reagent Chemicals

(Copies of this document are available online at http://pubs.acs.org/.)

ASTM INTERNATIONAL

ASTM D1193 - Standard Specification for Reagent Water

ASTM D1293 - Standard Test Methods for pH of Water

ASTM E200 - Standard Practice for Preparation, Standardization, and Storage of Standard and Reagent Solutions for Chemical Analysis

(Copies of these documents are available online at www.astm.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The sodium salts of EDTA shall be white, uniform, free flowing, and free from foreign matter.

a. Type I, the tetrasodium salt of EDTA, shall be $Na_4EDTA \cdot 4H_2O$ ($C_{10}H_{20}O_{12}N_2Na_4$).

b. Type II, the trisodium salt of EDTA, shall be $Na_3EDTA \cdot 3H_2O$ ($C_{10}H_{19}O_{11}N_2Na_3$).

c. Type III, the disodium salt of EDTA, shall be $Na_2EDTA \cdot 2H_2O$ ($C_{10}H_{18}O_{10}N_2Na_2$).

3.3 Composition. The material, as received, shall conform to the chemical characteristics in table I when tested as specified in table I.

TABLE I. Chemical requirements.

Test Parameter	Type I	Type II	Type III	Test
Chelating activity, oxalate method (mg $CaCO_3$/gram sample), minimum 1/	219	241	267	4.4.2
Chelating activity, potentiometric method (mg $CaCO_3$/gram sample) 1/	2/	2/	2/	4.4.3
Inflection point comparison (mL) 3/	±0.5	N/A	N/A	4.4.3.2.3
pH (1% solution) 4/	10.5 - 11.5	8.5 - 9.5	4.5 - 5.5	4.4.3.3
Water (%), maximum 5/	15.8	13.1	9.7	4.4.4
EDTA content, anhydrous (%), minimum 6/	83.2	86.2	89.3	4.4.5
Impurities (%), maximum	1.0	0.7	1.0	4.4.6

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TABLE I. Chemical requirements - Continued.

NOTES:

1/ Triplicate test results shall agree within ±5 milligrams (mg).

2/ The average chelating activity, potentiometric method shall agree within ±2.0 percent of the average determined by the oxalate method.

3/ For type I only, the volume of standard acid used to titrate to the second inflection point (volume 2) shall be twice the volume used to titrate to the first inflection point (volume 1) within ±0.5 milliliter (mL).

4/ Triplicate test results shall agree within ±0.2 pH unit.

5/ Triplicate test results shall agree within ±2.0 percent. The EDTA shall not char or change color to yellow or tan during the heating.

6/ Anhydrous EDTA, type I as Na_4EDTA, type II as Na_3EDTA, type III as Na_2EDTA.

3.4 Stability. The shelf life for all three types of EDTA shall be unlimited.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on a representative sample of the first lot produced when a first article sample is required (see 3.1). The inspection shall include the tests of 4.4.2 through 4.4.6, as specified in in table I.

4.3 Conformance inspection. Conformance inspection shall be performed in accordance with inspection provisions set forth herein. The characteristics set forth in table I, when tested in accordance with 4.4, shall constitute minimum inspections to be performed. Sampling shall be as specified (see 6.2). For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material submitted for inspection and delivery at one time. Failure of any test, by any sample, shall be cause for rejection of the lot represented.

4.4 Tests. The following tests shall be performed.

4.4.1 Quality assurance test methods. All tests shall be conducted in triplicate with the individual results and the average result recorded. Unless otherwise specified (see 6.2), all chemicals used in the test procedures shall be ACS reagent grade in accordance with ACS Reagent Chemicals. Water used in the test procedures shall be in accordance with ASTM D1193, type II reagent water. Chelating activity shall be determined by two methods, the oxalate method (see 4.4.2) and the potentiometric method (see 4.4.3). Both of these methods shall be run in triplicate.

4.4.2 Chelating activity, oxalate method.

4.4.2.1 Reagents for oxalate method.

4.4.2.1.1 Ammonium oxalate, saturated solution. Dissolve 60 grams of calcium-free ammonium oxalate in 1 liter of hot water and allow to cool to room temperature.

4.4.2.1.2 Sodium hydroxide, 1.0 normal (N) solution. Dissolve 40.0 grams of sodium hydroxide in 1 liter of water.

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4.4.2.1.3 Calcium chloride, standard solution (10 mg $CaCl_2$/mL). Transfer about 25 grams of calcium carbonate chelometric standard in accordance with ACS Reagent Chemicals to a platinum dish or crucible and dry at 200 to 210 °C for 4 hours. Cool in a desiccator. Add 10.01 grams of the dried calcium carbonate to 300 mL of water. Slowly and cautiously, add 18 mL of concentrated hydrochloric acid. Boil the resulting clear solution to remove carbon dioxide. After cooling to room temperature, add concentrated ammonium hydroxide until the solution is slightly alkaline to litmus, stir until dissolved, and dilute with distilled water to exactly 1 liter in a volumetric flask.

4.4.2.2 Test procedure for oxalate method (types I, II, and III). Weigh the test sample to 1.0000±0.0002 grams, then transfer into a beaker containing a polytetrafluoroethylene (PTFE)-coated stirring bar. Adjust to a pH of 11 to 12 by adding sodium hydroxide solution. Add 85 mL of water and 5 mL of saturated ammonium oxalate solution to the beaker and stir the mixture until the granular powder is completely dissolved. Titrate the sample at 25±3 °C with standard calcium chloride solution until the first permanent turbidity is detected. A black background and a light projected through the side of the solution aid in detecting the endpoint. Recheck the pH of the solution and, if necessary, readjust to a minimum pH of 11 by adding sodium hydroxide solution. If the turbidity disappears after the pH is adjusted, continue the titration until the turbidity is permanent. Record the volume (mL) of titrant required to reach the endpoint. Calculate the chelating activity, oxalate method (C_o) using the following equation:

$$C_o = \frac{(\text{mL } CaCl_2 \text{ solution}) \times 10}{\text{weight of sample (grams)}}$$

4.4.3 Chelating activity, potentiometric method.

4.4.3.1 Reagents for potentiometric method.

4.4.3.1.1 Sodium hydroxide, 0.1 N solution (type III). Prepare and standardize in accordance with ASTM E200. The values of three standardization titrations shall agree within ±0.0005 N. The value used shall be the average of the three determinations. The solution shall be standardized each day it is used.

4.4.3.1.2 Hydrochloric acid, 0.1 N solution (types I and II). Prepare and standardize in accordance with ASTM E200 using the sodium carbonate standardization procedure. The values of three standardization titrations shall agree within ±0.0009 N. The value used shall be the average of the three determinations. The solution shall be standardized weekly.

4.4.3.2 Test procedure for potentiometric method.

4.4.3.2.1 Sample preparation and pH (types I, II, and III). Weigh the test sample to 1.0000±0.0002 grams, then transfer into a beaker containing a PTFE-coated stirring bar. Add 100 mL of water and stir until the granular powder is completely dissolved. Determine the pH value of the stirred solution in accordance with ASTM D1293 Method A. Use the pH of the stirred solution for the pH value as required in table I and 4.4.3.3.

4.4.3.2.2 Potentiometric method (types I and II). Titrate the sample solutions prepared in 4.4.3.2.1 using standard 0.1 N hydrochloric acid and a 50-mL buret. Determine the volume of acid needed to reach the first volume inflection point (V_1) and the second volume inflection point (V_2) as follows:

a. In the vicinity of the inflection points, add l-mL increments of acid and record the pH or millivolt (mV) output of the pH measuring system. An automatic titrator also may be used to determine V_1 and V_2. Accumulate sufficient data to determine the inflection points by the second derivative method (refer to ACS Reagent Chemicals, “Second Derivative Method for the Determination of the Equivalence Point”). Interpolate volume 1 and volume 2 to 0.01 mL.

b. The chelating activity, potentiometric method (C_p) (mg $CaCO_3$ per gram of chelant) is calculated by the following equation:

$$C_p = \frac{(V_2 - V_1) \times \text{concentration of HCl (N)} \times 100.09}{\text{weight of sample (grams)}}$$

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4.4.3.2.3 Inflection point comparison (type I). Calculate the inflection point comparison using V_2 and V_1 from 4.4.3.2.2 by the following equation:

$$\text{Inflection point comparison (mL)} = V_2 - (2 \times V_1)$$

4.4.3.2.4 Potentiometric titration (type III). Titrate the sample solutions prepared in 4.4.3.2 using freshly prepared standard 0.1 N sodium hydroxide solutions in a 50-mL buret. Determine the volume of base needed to reach the inflection point as follows:

a. In the vicinity of the inflection point, add 1-mL increments of base and record the pH or mV output of the pH meter. An automatic titrator also may be used to determine the volume. Accumulate sufficient data to determine the inflection points by the second derivative method (refer to ACS Reagent Chemicals, "Second Derivative Method for the Determination of the Equivalence Point"). The volume shall be interpolated to 0.01 mL.

b. Calculate the chelating activity, potentiometric method (C_p) (mg $CaCO_3$ per gram of chelant) by the following equation:

$$C_p = \frac{\text{volume (mL)} \times \text{concentration (N) of NaOH} \times 100.09}{\text{weight of sample (grams)}}$$

4.4.3.3 pH of a 1-percent solution. Use the pH of the stirred solution in 4.4.3.2.1 for the pH value.

4.4.4 Water content (types I, II, and III). Transfer approximately 2.5 grams of the test sample into a clean, dry, tared weighing bottle (inner diameter 50 millimeters [mm] by height 30 mm) so that the sample is evenly spread over the bottom of the bottle, and measure the weight to an accuracy of ±0.2 mg. Place the weighing bottle (stopper ajar for water escape) and the sample in an oven set at 165±2 °C for type I EDTA, 110±2 °C for type II EDTA, and 140±2 °C for type III EDTA. Continue drying for 3 hours after the sample is put into the oven. Record the color of the sample after heating. While still hot, stopper the weighing bottle and allow it to cool in a desiccator over anhydrous calcium sulfate. Determine the weight of the dry sample (±0.2 mg) and calculate the percentage of water in the sample as follows:

$$\text{Water (\%)} = \frac{\text{((weight of original sample) - (weight of dried sample))} \times 100}{\text{weight of original sample}}$$

4.4.5 Anhydrous EDTA content (types I, II, and III). Calculate the anhydrous EDTA content from the chelating activity oxalate method (C_o) as follows:

Type I: $\text{EDTA content (as } Na_4\text{EDTA) (\%)} = C_o \times 0.3802$

Type II: $\text{EDTA content (as } Na_3\text{EDTA) (\%)} = C_o \times 0.3582$

Type III: $\text{EDTA content (as } Na_2\text{EDTA) (\%)} = C_o \times 0.3362$

4.4.6 Impurities (types I, II, and III). Calculate the impurities from the water content (see 4.4.4) and the anhydrous EDTA content (see 4.4.5) as follows:

$$\text{Impurities (\%)} = 100 - \text{Water} - \text{EDTA}$$

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sodium salts of EDTA covered by this specification are used for chemical treatment and cleaning of marine conventional fossil fuel boilers and for cleaning of marine distilling plants. The requirements and analytical methods in this specification were developed by Naval Research Laboratory Report 7689. Items acquired under this specification for military use are to be limited to the variety shown on the applicable military technical publication or military supply technical publication, including S9086-GX-STM-020/220, NSTM Chapter 220, Volume 2, Boiler Water/Feedwater Test And Treatment; S9531-BH-MMI-010, Heat Exchanger Repair of Heat Exchangers, Coolers, and Distilling Plants; S9221-C1-GTP-020, Main Propulsion Boilers; Repair & Overhaul; and Allowance Equipage Lists 2-560004118 and 2-560004107. Personnel of the military departments are requested to refer to these documents for guidance. Existing stocks of materials manufactured to MIL-T-24494B dated 19 October 1988 are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2).

c. When first article inspection is required (see 3.1).

d. Sampling requirements to an “accept at zero” (c=0) sampling plan, such as MIL-STD-1916 at verification level IV or ASQ H1331 at an acceptance quality limit (AQL) of 1.0 or other sampling plan as appropriate (see 4.3).

e. Requirements for the chemicals used in the test procedures, if not as specified (see 4.4.1).

f. Packaging requirements (see 5.1).

g. When safety data sheets are required (see 6.3).

6.3 Safety data sheets. When specified (see 6.2), contracting officers will identify those activities requiring copies of completed safety data sheets prepared in accordance with FED-STD-313.

6.4 National Stock Numbers (NSN). Table II provides a list of NSNs which correspond to this specification. This list may not include all possible NSNs associated with this document.

TABLE II. List of applicable NSNs.

NSN	Type	Unit of Issue	Nomenclature	Part Number
6810-01-109-3911	I	25-pound drum	Tetrasodium EDTA	M24494-I-025
6810-01-312-4076	II	25-pound drum	Trisodium EDTA	M24494-II-025
6810-01-450-6224	III	50-pound bag	Disodium EDTA	M24494-III-050

6.5 Alternate names. EDTA is also known commercially as edetic acid, (ethylenedinitrilo) tetraacetic acid, Hampene®, Sequestrene®, and Versene®.

6.6 Subject term (key word) listing.

Chelating activity

EDTA

Organic and inorganic impurities

Oxalate

Potentiometer titration

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
Navy – SH
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